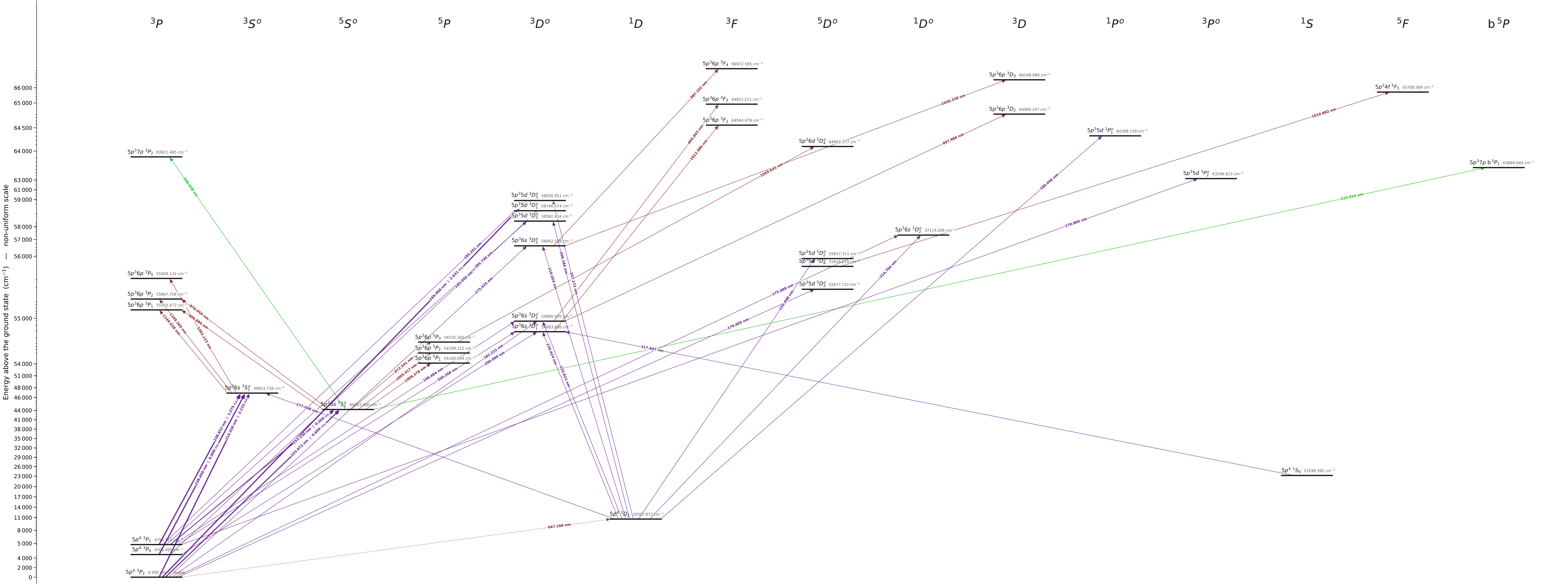
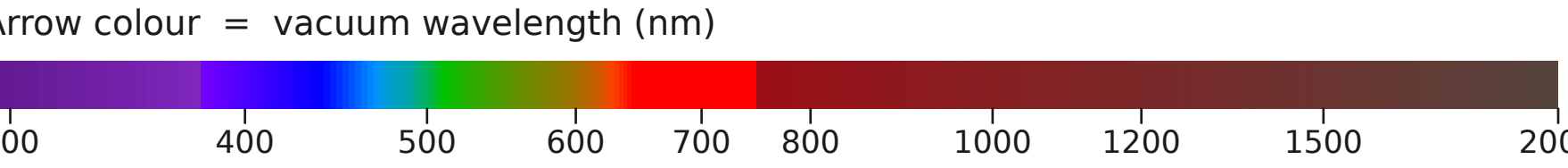


Tellurium

Te (Te I). Strongest classified transitions below 2 μm (NIST): 34 levels, 45 lines.

Te⁺ ionisation limit 72669.01 cm⁻¹ = 9.00981 eV (137.610 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element |(J||er||J')| in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
5p ⁴ ³ P ₂ – 5p ³ 5d ³ D ₃ ^o	169.9901	–	1763.5876	2.631	4.080e+08	–	–	
5p ⁴ ³ P ₂ – 5p ³ 6s ³ S ₁ ^o	214.3497	214.2822	1398.6139	2.133	3.120e+08	–	–	
5p ⁴ ³ P ₂ – 5p ³ 6s ⁵ S ₂ ^o	225.9734	225.9034	1326.6716	0.604	1.280e+07	–	–	
5p ⁴ ³ P ₀ – 5p ³ 6s ³ S ₁ ^o	238.4004	238.3277	1257.5167	0.900	4.040e+07	–	–	
5p ⁴ ³ P ₁ – 5p ³ 6s ³ S ₁ ^o	238.6519	238.5792	1256.1911	1.275	8.080e+07	–	–	
5p ⁴ ³ P ₁ – 5p ³ 6s ⁵ S ₂ ^o	253.1499	253.0738	1184.2488	0.209	1.090e+06	–	–	

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).